Graphics Study: Comparing Two Simulator Renderers

A small user study comparing how the same virtual reality temporal bone drilling task looks and feels with two different graphics renderers. Your performance is not being evaluated.

What you will do (about 40–50 minutes)

1. **Tutorial.** In the headset, a short guided tutorial on the drill, haptic device, microscope, zoom, foot pedals and pause menu.
2. **Session 1.** Up to 10 minutes of drilling: perform a cortical mastoidectomy and identify the short process of the incus. Tell the proctor when you have found it. If you finish early, hold the left foot pedal and select **End trial**.
3. **Workload questionnaire (NASA-TLX)** on the desktop, headset off.
4. **Session 2.** The same task with the other renderer (no tutorial), followed by the workload questionnaire again.
5. **Final questions.** A brief background questionnaire, which session looked more realistic and which you would rather use to train, and a short reflection.

The anatomy, controls, headset view and task are identical in both sessions; only the graphics differ. The order of the two renderers is randomised and balanced across participants.